

Daily Progress Report

Week 6_Day1: Form Handling & Controlled Inputs

Program	3-Month MERN Stack Internship Program
Intern Name	Mohsin Ali
Week	Week_6 (Day 1)
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Day 1: Form Handling & Controlled Inputs

1.1 Topics Studied

- Understanding the difference between Controlled and Uncontrolled Components in React
- Managing form data using controlled components
- Using a single state object to manage multiple form fields efficiently
- Handling user input with the onChange event
- Updating object state using the spread operator (...)
- Implementing real-time form validation
- Validating email addresses using regular expressions (Regex)
- Enabling and disabling the submit button based on form validity
- Displaying validation error messages using conditional rendering
- Handling form submission using the onSubmit event
- Resetting form fields after successful submission
- Building responsive and reusable form components using modern React practices

1.2 Task Description

Developed a comprehensive **"Contact Us"** form component using React that demonstrates the implementation of controlled components and centralized form state management. All form inputs, including **Full Name**, **Email**, **Subject**, and **Message**, are fully controlled through a single React state object. Real-time validation was implemented to provide immediate feedback as users type, ensuring required fields are completed correctly and that the email follows a valid format. The submit button remains disabled until the entire form satisfies all validation rules, preventing invalid submissions. Upon successful submission, the entered data is processed, the form is reset, and a confirmation message is displayed. The

application was organized into reusable components with responsive styling to provide a clean and user-friendly interface.

1.3 Implementation Details

- Created a reusable `ContactForm` component.
- Managed all form fields using a single `useState` object.
- Implemented controlled inputs by binding each field's value to React state.
- Updated form data dynamically using a single `handleChange` function.
- Used the spread operator to update individual properties while preserving the remaining state.
- Implemented real-time validation for all input fields.
- Validated email addresses using a Regular Expression (Regex).
- Displayed validation messages dynamically using conditional rendering.
- Disabled the Send Message button until all validation rules were satisfied.
- Handled form submission using the `onSubmit` event with `preventDefault()`.
- Displayed a success message after successful submission.
- Reset all form fields to their initial state after successful submission.
- Organized the project into separate component and styling files to maintain a clean and scalable project structure.
- Applied responsive CSS to ensure the form adapts smoothly across desktop and mobile devices.

1.4 Outcome

Successfully developed a professional **Contact Us** form that demonstrates the practical implementation of controlled components in React. The project strengthened understanding of managing complex form data using a single state object, handling user input through controlled fields, implementing real-time validation, and improving user experience by preventing invalid form submissions. Additionally, organizing the application into reusable components and maintaining responsive styling reinforced best practices for building scalable and maintainable React applications.

Screenshots:

```
1 import { useState } from "react";
2 import "../ContactForm.css";
3
4 function ContactForm() {
5   const [formData, setFormData] = useState({
6     fullName: "",
7     email: "",
8     subject: "",
9     message: ""
10  });
11
12  const [submitted, setSubmitted] = useState(false);
13
14  const emailRegex = /^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}$/;
15
16  const errors = {
17    fullName:
18      formData.fullName.trim().length < 3
19      ? "Enter at least 3 characters."
20      : "",
21    email:
22      formData.email === ""
23      ? "Email is required."
24      : !emailRegex.test(formData.email)
25      ? "Enter a valid email address."
26      : "",
27    subject:
28      formData.subject.trim().length < 3
29      ? "Enter at least 3 characters."
30      : ""
31  };
32
33  const handleSubmit = (e) => {
34    e.preventDefault();
35    if (errors.fullName || errors.email || errors.subject) return;
36    setSubmitted(true);
37  };
38
39  return (
40    <div>
41      <div className="form-group">
42        <label>Full Name</label>
43        <input
44          type="text"
45          name="fullName"
46          value={formData.fullName}
47          onChange={handleChange}
48          placeholder="Enter your full name"
49        />
50        {errors.fullName && <small>{errors.fullName}</small>}
51      </div>
52      <div className="form-group">
53        <label>Email</label>
54        <input
55          type="text"
56          name="email"
57          value={formData.email}
58          onChange={handleChange}
59          placeholder="Enter your email"
60        />
61        {errors.email && <small>{errors.email}</small>}
62      </div>
63      <div className="form-group">
64        <label>Subject</label>
65        <input
66          type="text"
67          name="subject"
68          value={formData.subject}
69          onChange={handleChange}
70          placeholder="Enter subject"
71        />
72        {errors.subject && <small>{errors.subject}</small>}
73      </div>
74      <div className="form-group">
75        <label>Message</label>
76        <textarea
77          name="message"
78          value={formData.message}
79          onChange={handleChange}
80          placeholder="Write your message"
81        />
82        {errors.message && <small>{errors.message}</small>}
83      </div>
84      <div className="button">
85        <button type="button" disabled={!isFormValid} onClick={handleSubmit}>
86          Send Message
87        </button>
88      </div>
89    </div>
90  );
91}
```

```
4 function ContactForm() {
5   // ... (JavaScript code from previous screenshot)
6
7   return (
8     <div>
9       <div className="form-group">
10        <label>Full Name</label>
11        <input
12          type="text"
13          name="fullName"
14          value={formData.fullName}
15          onChange={handleChange}
16          placeholder="Enter your full name"
17        />
18        {errors.fullName && <small>{errors.fullName}</small>}
19      </div>
20      <div className="form-group">
21        <label>Email</label>
22        <input
23          type="text"
24          name="email"
25          value={formData.email}
26          onChange={handleChange}
27          placeholder="Enter your email"
28        />
29        {errors.email && <small>{errors.email}</small>}
30      </div>
31      <div className="form-group">
32        <label>Subject</label>
33        <input
34          type="text"
35          name="subject"
36          value={formData.subject}
37          onChange={handleChange}
38          placeholder="Enter subject"
39        />
40        {errors.subject && <small>{errors.subject}</small>}
41      </div>
42      <div className="form-group">
43        <label>Message</label>
44        <textarea
45          name="message"
46          value={formData.message}
47          onChange={handleChange}
48          placeholder="Write your message"
49        />
50        {errors.message && <small>{errors.message}</small>}
51      </div>
52      <div className="button">
53        <button type="button" disabled={!isFormValid} onClick={handleSubmit}>
54          Send Message
55        </button>
56      </div>
57    </div>
58  );
59}
```

Screenshots:

This screenshot shows a contact form with a blue sidebar on the left and a white main area on the right. The sidebar contains the heading 'Contact Us' and a list of three bullet points: '✓ Fast Response', '✓ Professional Support', and '✓ Secure Communication'. The main area contains four input fields: 'Full Name' (with value 'John Smith' and error 'Enter at least 3 characters.'), 'Email' (with value 'john@example.com' and error 'Email is required.'), 'Subject' (with value 'How can we help?' and error 'Subject should contain at least 5 characters.'), and 'Message' (with placeholder 'Write your message...'). A vertical scrollbar is visible on the right side of the form.

Contact Us

Have questions, suggestions, or need assistance?
Fill out the form and our team will get back to you within 24 hours.

- ✓ Fast Response
- ✓ Professional Support
- ✓ Secure Communication

Full Name
John Smith
Enter at least 3 characters.

Email
john@example.com
Email is required.

Subject
How can we help?
Subject should contain at least 5 characters.

Message
Write your message...

This screenshot shows the same contact form as above, but with different input values. The 'Email' field now contains 'mohsinalirana222gmail.com' with the error 'Enter a valid email address.'. The 'Subject' field contains 'hello'. The 'Message' field contains 'Hello, My name is Mohsin Ali.' and has a character count of '30/500' at the bottom right. A 'Send Message' button is located at the bottom of the form. A vertical scrollbar is visible on the right side of the form.

Contact Us

Have questions, suggestions, or need assistance?
Fill out the form and our team will get back to you within 24 hours.

- ✓ Fast Response
- ✓ Professional Support
- ✓ Secure Communication

Email
mohsinalirana222gmail.com
Enter a valid email address.

Subject
hello

Message
Hello, My name is Mohsin Ali.
30/500

Send Message

Contact Us

Have questions, suggestions, or need assistance?
Fill out the form and our team will get back to you
within 24 hours.

- ✓ Fast Response
- ✓ Professional Support
- ✓ Secure Communication

Email

mohsinalirana222@gmail.com

Subject

hello

Message

Hello, My name is Mohsin Ali.

30/500

Send Message